

AF-4500 Series Centrifugal Pump

Installation, Operation & Maintenance Manual

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Applies to models: AF-4510, AF-4520, AF-4530

1. Overview & Scope

The AF-4500 series is a range of horizontal, single-stage, end-suction centrifugal pumps designed for continuous-duty service in water treatment, chemical processing, and general industrial transfer applications. The series covers flow rates up to 850 m³/h and heads up to 120 m. This manual covers receiving, installation, commissioning, operation, and routine maintenance for all AF-4500 series models. It should be read in full by installation and maintenance personnel before first use.

2. Safety Precautions

- Always isolate and lock out electrical supply (LOTO) before opening the pump casing or bearing housing.
- Allow the pump casing to cool to ambient temperature before servicing if handling hot fluids.
- Wear safety glasses, gloves, and steel-toe boots during all installation and maintenance work.
- Never operate the pump against a fully closed discharge valve for more than 60 seconds.
- Verify the pumped fluid is compatible with wetted-part materials before startup.

3. Technical Specifications

Model	Max Flow (m³/h)	Max Head (m)	Motor Power (kW)	Inlet / Outlet (mm)	Weight (kg)	Operating Temp Range (°C)	Max Working Pressure (bar)
AF-4510	450	95	55	150 / 125	380	-10 to 120	16
AF-4520	650	110	90	200 / 150	520	-10 to 120	16
AF-4530	850	120	132	250 / 200	710	-10 to 120	25

4. Installation Procedure

- 1 Inspect the pump and baseplate for shipping damage before installation; report damage within 48 hours.
- 2 Pour and cure a reinforced concrete foundation sized to at least 1.5x the combined pump/motor weight.
- 3 Set the baseplate on the foundation using shims; check level to within 0.5 mm/m in both axes.
- 4 Grout the baseplate and allow to cure per grout manufacturer's instructions (minimum 48 hours) before alignment.
- 5 Align pump and motor shafts using dial indicators; angular and parallel misalignment must not exceed 0.05 mm.
- 6 Connect suction and discharge piping independently supported — piping must never impose load on the pump nozzles.
- 7 Complete electrical connections per the motor nameplate and verify rotation direction with the coupling disconnected.
- 8 Perform the pre-start checklist in Section 5.1 before commissioning.

5. Operation

5.1 Pre-Start Checklist

- Suction and discharge valves in correct position and piping fully vented.
- Bearing housing filled with Atlas HydraSeal AH-220 to the sight-glass level.
- Coupling guard installed and secured.
- Motor rotation direction confirmed to match the direction arrow on the casing.
- Pump casing primed (flooded) — never dry-run the pump.

5.2 Startup

Open the suction valve fully, open the discharge valve slightly, start the motor, and open the discharge valve fully once stable flow is established. Monitor discharge pressure, motor amperage, and bearing housing temperature during the first 30 minutes of operation.

5.3 Normal Operation Monitoring Points

- Bearing housing temperature: should stabilize below 80°C.
- Vibration at bearing housing: should remain below 4.5 mm/s RMS.
- Seal chamber: no visible leakage beyond normal weeping during break-in (first 24 hours only).

5.4 Shutdown

Close the discharge valve gradually, stop the motor, then close the suction valve if the pump will be idle for an extended period. For freezing conditions, drain the casing fully.

6. Maintenance Schedule

Interval	Task	Details
Daily	Visual inspection	Check for leaks at seal, casing joints, and piping connections.
Weekly	Vibration check	Record bearing housing vibration; compare to baseline.
Monthly	Coupling inspection	Check coupling alignment and element wear; re-torque guard bolts.
Quarterly	Bearing lubrication	Re-grease bearings using Atlas HydraSeal AH-220 only.
Annually	Mechanical seal service	Inspect and replace mechanical seal per SOP-114.
Annually	Wear ring inspection	Measure impeller wear ring clearance; replace if beyond limit (Section 7).

7. Impeller Wear Ring Clearance

Excessive wear ring clearance is the leading cause of reduced flow and head in AF-4500 series pumps. Measure radial clearance at each annual service and replace wear rings when clearance exceeds the maximum wear limit below.

Model	New Clearance (mm)	Max Wear Limit (mm)
AF-4510	0.35 – 0.40	0.75
AF-4520	0.40 – 0.45	0.75
AF-4530	0.45 – 0.50	0.80

8. Troubleshooting Quick Reference

For detailed troubleshooting steps, refer to the companion document *AF-4500 Troubleshooting Guide*. Common issues include reduced flow (check wear rings, priming, rotation), excessive noise (check for cavitation), and bearing overheating (check lubrication level and type).

Appendix A — Casing Bolt Torque Specifications

Model	Bolt Size	Torque (Nm)	Tightening Pattern
AF-4510	M12	60	Star pattern, 2 passes
AF-4520	M16	95	Star pattern, 2 passes
AF-4530	M20	140	Star pattern, 3 passes

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